

Polynomially conditioned primitives in separated Legendre bands

Proof independently reviewed by root, size_bounds, and independent_directions

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All unqualified norms use Lebesgue measure on $[0, 1]$. Write

$$w(t) = t(1 - t), \quad \lambda_d = d(d + 1), \quad H_d(t) = \int_0^t P_d(2u - 1) du.$$

This note gives analytic tools for ordered-prefix moment constraints. It does not assert independence of every family of such constraints, or a construction of three simultaneous small dice.

Lemma 1 (An exact primitive formula). *Fix $d \geq 2$ and $0 \leq r \leq (d - 2)/2$. The polynomial operator*

$$\mathcal{L}_d C = (wC)'' + \lambda_d C$$

is invertible on $\mathcal{P}_r$. For $Q \in \mathcal{P}_r$, let

$$C_1 = \mathcal{L}_d^{-1} Q', \quad C_0 = -\mathcal{L}_d^{-1} Q, \quad B_a = wC_a.$$

Then, identically on $[0, 1]$,

$$\int_0^t Q(u) H_d'(u) du = (Q - B_1') H_d + B_1 H_d', \quad (1)$$

$$\int_0^t Q(u) H_d(u) du = -B_0' H_d + B_0 H_d'. \quad (2)$$

The coefficient polynomials have degrees at most $r + 2$ and norms at most $C(r + 1)^5 \|Q\|_2$, with an absolute constant.

Proof. The shifted Legendre differential equation gives $H_d'' = -\lambda_d H_d / w$ in $(0, 1)$. For arbitrary $A, B = wC$,

$$(AH_d + BH_d')' = (A' - \lambda_d C)H_d + (A + B')H_d'.$$

The two choices in the statement give the desired derivatives. Both right sides vanish at zero because $H_d(0) = B_a(0) = 0$.

For invertibility, use the shifted Jacobi polynomials with parameters $(1, 1)$, orthogonal for the weight w . The polynomial of degree j satisfies

$$wJ_j'' + 2w'J_j' + j(j + 3)J_j = 0,$$

so it is an eigenvector of $\mathcal{L}_d$ with eigenvalue $\lambda_d - (j + 1)(j + 2)$. For $j \leq r \leq (d - 2)/2$, these eigenvalues are at least cd^2 . Consequently

$$\|\mathcal{L}_d^{-1} F\|_{L^2(w)} \leq Cd^{-2} \|F\|_{L^2(w)}.$$

For a polynomial C of degree at most r , $\|C\|_2 \leq C(r + 1)\|C\|_{L^2(w)}$. Indeed, remove intervals of length $c(r + 1)^{-2}$ at the endpoints and use $\|C\|_\infty \leq (r + 1)\|C\|_2$ to retain at least half its squared norm; on the remainder $w \geq c(r + 1)^{-2}$. The polynomial L^2 Markov inequality $\|Q'\|_2 \leq C(r + 1)^2\|Q\|_2$ now bounds C_1 by $C(r + 1)^3 d^{-2}\|Q\|_2$ and C_0 by $C(r + 1)d^{-2}\|Q\|_2$. Applying the same derivative estimate to wC_a proves the asserted, deliberately coarse coefficient bound. $\square$

Lemma 2 (Three separated bands). *Suppose $d > 2r + 1$, $p, q, v \in \mathcal{P}_r$, and $D \geq d + r + 1$. Then*

$$(\|p\|_2^2 + \|q\|_2^2 + \|v\|_2^2)^{1/2} \leq C(D + 1)^6 \|p + H_d q + H'_d v\|_2.$$

If H_d, H'_d are both multiplied by $0 < \eta \leq 1$, the bound costs at most a further factor η^{-1} .

Proof. Put $G = H_d q + H'_d v$. Since H_d is a linear combination of shifted Legendre polynomials of degrees $d - 1, d + 1$, and H'_d has degree d and is orthogonal to $\mathcal{P}_{d-1}$, we have $G \perp \mathcal{P}_r$. For example, testing $H_d q$ against $p \in \mathcal{P}_r$ reduces to orthogonality of H_d against a polynomial of degree at most $2r < d - 1$. Thus $\|p + G\|_2^2 = \|p\|_2^2 + \|G\|_2^2$.

At the $d + 1$ zeros t_j of H_d , including the endpoints, $G(t_j) = H'_d(t_j)v(t_j)$. The positive Gauss–Lobatto weights are

$$\omega_j = \frac{1}{d(d+1)H'_d(t_j)^2}, \quad \sum_j \omega_j = 1,$$

and this rule integrates polynomials through degree $2d - 1$. In particular $|H'_d(t_j)| \geq [d(d+1)]^{-1/2}$ and

$$\|v\|_2^2 = \sum_j \omega_j v(t_j)^2 \leq d(d+1) \|G\|_\infty^2 \leq C(D+1)^4 \|G\|_2^2.$$

Here $\deg G \leq D$ and the last step is the polynomial evaluation bound. Since $\|H'_d\|_\infty \leq 1$, $\|H_d q\|_2 \leq \|G\|_2 + \|v\|_2$. The reviewed multiplication estimate $\|H_d q\|_2 \geq c(D+1)^{-4} \|q\|_2$ from `lacunary_legendre_conditioning.tex` completes the proof. Scaling is immediate. $\square$

Theorem 1 (A fixed number of trinary bands). *Fix k . Let $r \geq 0$, $0 < \eta_i \leq 1$, and positive integers d_i satisfy*

$$d_i > 2 \left(r + \sum_{j < i} (d_j + 1) \right) + 1.$$

Set $D = r + \sum_{i=1}^k (d_i + 1)$ and $Z_{i,0} = 1$, $Z_{i,1} = \eta_i H_{d_i}$, $Z_{i,2} = \eta_i H'_{d_i}$. For arbitrary $p_\alpha \in \mathcal{P}_r$, $\alpha \in \{0, 1, 2\}^k$,

$$\left(\sum_\alpha \|p_\alpha\|_2^2 \right)^{1/2} \leq C_k (D + 1)^{6k} \prod_{i=1}^k \eta_i^{-1} \left\| \sum_\alpha p_\alpha \prod_{i=1}^k Z_{i,\alpha_i} \right\|_2.$$

Minimal admissible choices have $D = O_k(r + 1)$. Evaluation on any node array whose star discrepancy is at most $c(D + 1)^{-2}$ preserves this lower bound up to an absolute constant.

Proof. Group the sum by its last coordinate. Each of its three coefficients has degree at most $r + \sum_{j < k} (d_j + 1)$, so Lemma 2 applies. Iterate, summing the resulting squared inequalities. The minimal degree recurrence grows by a factor at most three per step. For sampling, every resulting polynomial F has degree at most D , and integration by parts for the empirical distribution function gives

$$\left| K^{-1} \sum_q F(b_q)^2 - \int_0^1 F^2 \right| \leq \text{disc}(b) \int_0^1 |(F^2)'| \leq C(D + 1)^2 \text{disc}(b) \|F\|_2^2.$$

Choose the absolute discrepancy constant small enough. $\square$

Use and limitation. An ordered prefix involving a new highest-frequency density contains terms $\int QH'_d$ or $\int QH_d$, where the previous factors form a lower-degree polynomial Q . Lemma 1 represents such a term in two adjacent high-frequency bands with polynomially controlled coefficients. Theorem 1 separates explicit trinary product families. It does not automatically separate arbitrary ordered-prefix families: after applying $\mathcal{L}_d^{-1}$, coefficient polynomials need not retain a prescribed product form. Independence and positive completion for those larger families remain separate questions.